MIL-S-43897B
5 December 1983
SUPERSEDING
MIL-S-43897A
25 April 1977

MILITARY SPECIFICATION

SHOES, ELECTRICAL HAZARDS PROTECTIVE, OXFORD, MEN'S

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers black, blucher, oxford, safety, electrical hazard protective shoes with traction tread soles and heels, and steel box-toes.

1.2 Classification. The shoes shall be of one type in the following sizes and widths, as specified (see 6.2).

TABLE I. Schedule of sizes

Schedule of widths	4	4½	5	5½	6	6½	7	7½	8	8½	9	9½	10	10½	11	11½	12	12½	13	13½	14
XN	-	-	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
N	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
R	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
XW	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research and Development Center, Natick, MA 01760 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter

FSC 8430

2. APPLICABLE DOCUMENTS

2.1 Government documents. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

SPECIFICATIONS

FEDERAL

C-F-206	-	Felt Sheet: Cloth, Felt, Wool, Pressed.
V-L-61	-	Laces, Nylon.
V-T-285	-	Thread, Polyester.
V-T-295	-	Thread, Nylon.
KK-I-570	-	Insole, Footwear, Leather, Cattlehide.
TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings.
CCC-C-443	-	Cloth, Duck, Cotton (Single and Plied Filling Yarns, Flat).
PPP-B-566	-	Boxes, Folding, Paperboard.
PPP-B-636	-	Boxes, Shipping, Fiberboard.
PPP-B-676	-	Boxes, Setup.

MILITARY

MIL-L-3122	-	Leather, Cattlehide, for Footwear Uppers, Chrome Tanned, Fatliquored.
MIL-S-22777	-	Soles and Heels, Rubber, Traction Tread, Shoe.
MIL-C-41814	-	Counter, Footwear.
MIL-L-43585	-	Lasts, Footwear, Shoe, Safety Toe, Men's, U.S. MIL-7.

STANDARDS

FEDERAL

FED-STD-191	-	Textile Test Methods.
FED-STD-311	-	Leather, Methods of Sampling and Testing.
FED-STD-406	-	Plastics: Methods of Testing.
FED-STD-751	-	Stitches, Seams, and Stitchings.

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.

(Copies of documents required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Other publications. Unless otherwise specified, the following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this document to the extent specified herein:

AMERICAN SOCIETY FOR TESTING AND MATERIALS STANDARDS (ASTM)

D 5	-	Penetration of Bituminous Materials.
D 412	-	Rubber Properties in Tension.
D 573	-	Deterioration in an Air Oven.
D 746	-	Brittleness Temperature of Plastics and Elastomers by Impact.
D 1052	-	Rubber Deterioration - Cut Growth Using Ross Flexing Apparatus.
D 1238	-	Flow Rates of Thermoplastics by Extrusion Plastometer.
D 2240	-	Rubber Property - Durometer Hardness
D 3951	-	Standard Practice for Commercial Packaging.
E 18	-	Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials.
E 28	-	Softening Point by Ring-and-Ball Apparatus.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

Z41.1 American National Standard for Men's Safety Toe Footwear

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variation from this document may appear in the sample, in which case this document shall govern.

3.2 First article. When specified a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.4).

3.3 Materials (see 6.6).

3.3.1 Leather. All leather components shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide.

3.3.1.1 Upper leather vamps, quarters, and tongues. The leather for the vamps, quarters, and tongues shall be smooth plated, full grain or corrected grain side leather conforming to treatment B of MIL-L-3122 except that: the tearing strength (slot tear) shall not apply, the leather shall be treated with an approved water resistant compound, the maximum amount of chloroform soluble material permitted in this leather shall be 18.0 percent, and the thickness shall be as specified for the cut parts in 3.7.1. The break for vamps and quarters shall not be more than a break pattern of 4 when examined as specified in 4.4.3. The color shall be black. All finishes applied by the shoe manufacturer and tanner shall be compatible with each other to insure a long lasting lustrous finish. When tested as specified in 4.4.1, at least 80 percent of the specimens tested shall have minimum stitch tearing strength values of 30 pounds.

* 3.3.1.2 Insoles. The insoles shall be cut from leather conforming to type I, tannage a or b, class 1 of KK-I-570. The insoles shall have either a channeled or stuck-on rib (see 3.3.5.3). The thickness of the channeled insoles shall be 6-1/2 irons and the thickness of the stuck-on rib insoles shall be 5-1/2 irons. As an alternate, the insole may be cut from Texorist material. The Texorist shall contain 0.25 to 1.00 percent copper-8-quinolinolate uniformly distributed throughout the insole. The insole shall be 0.116 $\pm$ 0.010 inch thick. The insole shall be cut with the heel-to-toe direction across the machine direction of the Texorist. The Texorist insole shall have a stuck-on rib applied to the printed side of the material (see 3.3.5.3).

* 3.3.1.3 Vamp and quarter linings. The vamp and quarter linings shall be cut from soft pliable calf, kip, or side leather 3 + 1/2 ounces in thickness when tested as specified in 4.4.1. The leather shall be either chrome or vegetable tanned or a combination thereof. The leather shall be full grain or partially corrected grain. The color of the leather shall be as produced by the tanning agents. A light application of dye may be used to level the color of the grain surface. As an alternate to the above, the lining material may be vegetable tanned, chrome tanned, or vegetable chrome retanned split pigskin. The pigskin shall be the first split from the grain side after the grain has been removed. The thickness shall be 2 to 3 ounces. The pigskin shall be drum dyed and the color shall be light russet. The surface of the split that was nearest the grain surface shall be sealed with a clear lacquer and then smooth plated. The sealed plated surface shall be worn next to the foot. All the lining leather shall contain 0.18 to 0.70 percent paranitrophenol when tested as specified in 4.4.1. Slaughter cuts and flanky or pipey leather which may effect serviceability will not be permitted.

* 3.3.1.4 Heel pads. The heel pads shall be cut from either chrome tanned or a combination of chrome and vegetable tanned, full grain or partially corrected lining leather, sheepskin, kip, calf, or cattlehide. The heel pads shall have a minimum thickness of 2 ounces and a maximum thickness of 3-1/2 ounces when tested as specified in 4.4.1. The color of the leather shall be as produced by the tanning agents. A light application of dye may be used to level the color of the grain surface. Heel pads cut from lining leather specified in 3.3.1.3 may be used. The heel pad shall contain not less than 0.18 percent nor more than 0.70 percent paranitrophenol fungicide when tested as specified in 4.4.1. As an alternate to the above, the heel pads may be cut from the pigskin lining leather specified in 3.3.1.3.

3.3.2 Counters. The counters shall be leather, leatherboard, shoeboard, or polyethylene.

3.3.2.1 Leatherboard or shoeboard counters. The leatherboard, or shoeboard counters shall conform to MIL-C-41814 as specified for shoes, electrical hazards, protective oxford. The counters shall be full molded conforming to the MIL-7 last and the bottom flange shall be molded to provide for welt construction completely around the heel seat periphery of the shoe.

3.3.2.2 Polyethylene counters. The counters shall be molded and the bottom flanged as specified in 3.3.2.1 from an unfilled polyethylene resin of natural color. The molded pieces shall be uniform in texture and finish and shall be free from porosity, cracks, and blisters. The polyethylene resin shall conform to the requirements of table II when tested as specified in 4.4.1.

TABLE II. Polyethylene resin requirements

Characteristic	Requirement
Density	0.91 to 0.93 grams per milliliter
Melt index	1.0 to 5.0 grams per 10 minutes
Tensile strength, ultimate	1400 psi minimum
Elongation, ultimate	400 percent minimum

3.3.3 Rubber compounds.

3.3.3.1 Outsole. The outsole shall conform to type I, grade B of MIL-S-22777 except that the color may be mahogany brown.

3.3.3.2 Heel, rubber. The rubber heel shall conform to type II, grade B, class 2 of MIL-S-22777 except that the color may be mahogany brown.

3.3.3.3 Midsole. The midsole shall be of non-marking rubber. The hardness of the midsole shall be not less than 88 or more than 95 before and after aging when determined by the Shore A Durometer. The cut growth after aging shall not exceed 500 percent after being subjected to 25,000 flexes when tested as specified in 4.4.1. The color shall approximately match the color of the outsole. The thickness of the midsole shall be not less than 5 irons or more than 6 irons.

3.3.3.4 Welting, vinyl. The welting shall be vinyl, bright flat top with the construction providing for a storm welt of medium bead number 438 or any other bead number of equal performance. The welting shall be of sufficient width to assure that the required extensions on the finished shoes are met (see 3.7.16). The thickness of the welting shall be 0.12 to 0.13 inch. The base compound shall be virgin polyvinyl chloride, compounded with suitable non-bleeding pasticizers. The finished welting shall be smooth, uniform in texture, and free from porosity, blemishes, cracks, blisters, and other defects affecting serviceability and appearance. The vinyl welting shall conform to the requirements of table III when tested as specified in 4.4.1.

TABLE III. Vinyl welting requirements

Characteristic	Requirement
Hardness	85-95
Specific gravity	1.32 maximum
Brittle point	-15°F or less
Brittle point, after aging	+20°F maximum change from standard brittle point
Tensile strength, psi	2300 minimum
Elongation, percent	225 minimum
Water absorption, percent gain	2.20 maximum

3.3.4 Metals.

3.3.4.1 Steel box toe. The steel box toe shall be fabricated from cold-rolled carbon steel and shall conform, after heat treatment, to the requirements of table IV when tested as specified in 4.4.1.

* TABLE IV. Physical requirements, steel box toe

Hardness (Rockwell C scale)	Carbon content (percent)	Thickness inch	Sizes	Shape	Trade pattern	Plating and coating
43 to 50	0.50 to 0.82	0.062 ± 0.0025	8	Conform to toe of last	400	Zinc electro-plate or zinc com-pound coating 1/

1/ The steel box toes shall be thoroughly cleaned by any method of TT-C-490 prior to coating. The zinc compound coating shall be completely and uniformly applied to the base metal.

3.3.4.1.1 Impact resistance. The steel box toe of the finished shoe shall have a minimum inside clearance of 1/2 inch when tested for impact resistance as specified in 4.5.4.

3.3.5 Fabrics.

3.3.5.1 Channeled insole reinforcement. The insole reinforcement shall be a cotton duck cloth conforming to type I of CCC-C-443.

3.3.5.2 Felt strip, masking tape, and toe cushion assembly.

3.3.5.2.1 Felt strip. The felt strip for masking the steel toe breast line shall be a minimum of 1/2 inch in width and shall conform to type III, classification 9A-2 of C-F-206 except that the minimum breaking strength shall be 20 pounds.

3.3.5.2.2 Masking tape. The masking tape for holding the felt strip against the breast line of steel box toe shall be 1-1/2 inch wide pressure sensitive adhesive coated cloth tape.

3.3.5.2.3 Toe cushion. The toe cushion to cover the steel box toe and cover the masking tape area shall be cut from a 1/8 inch thick polyurethane pad.

3.3.5.3 Stuck-on rib (when used, see 3.3.1.2). The finished rib shall be 7/32 to 8/32 inch high, not less than 5/8 inch wide measured from the inside vertical portion of the rib, and shall extend around the periphery of the insole to provide for a welted heel seat. The rib shall be made from coated fabric or a combination of coated fabric and fiberboard. The fabric shall conform to requirements specified in 3.3.5.3.1. The upstanding portion of the rib shall be one of the following: Not less than three plies of coated fabric folded over on itself; two layers of coated fabric with a fiberboard filler not less than 0.050 inch thickness; or fiberboard and coated fabric.

* 3.3.5.3.1 Rib construction. The ribbing fabric used in the construction of the rib shall be unbleached cotton or cotton-synthetic fabric conforming to the requirements of table V. The fabric shall be coated on one side with a rubberbased adhesive specified in 3.3.10 and bonded to the flesh side of the insole. The fabric or fiberboard shall cover 5/32 -2/32, +1/32 inch of the peripheral edge and extend over the top of the vertical rib, or the fabric may extend from the inside top of the vertical rib and cover the area between the ribs for a distance not less than 5/8 inch from the inside rib edge. The preformed rib shall be firmly and smoothly fitted and bonded to the flesh side of the insole as specified above to provide for a welted heel seat with the required edge extension on the finished shoe. As an alternate, the fabric bonded to the flesh side of the insole between the ribs may have a pinked edge.

TABLE V. Ribbing fabric requirements

Weight, ounces per sq. yd. (min .)	Yarns per inch (min.)		Breaking strength (min.), pounds, warp and filling
	Warp	Filling	
8.0	52	30	105

3.3.5.3.2 Rib strength. The rib strength shall conform to table VI when tested as specified in 4.4.2.3.

TABLE VI. Rib strength requirements

Characteristic	Minimum 1/	Average
Shear strength	70 lbs.	75 lbs.
Stitch strength	20 lbs.	30 lbs.

1/ No single determination (sample unit) shall fall below the minimum value specified and the average of all specimens (lot average) shall not be less than the average specified.

3.3.6 Thread.

3.3.6.1 Thread, upper fitting. All upper fitting operations shall be stitched with nylon thread conforming to type I, II, or III, class A of V-T-295, except that requirements for colorfastness shall not apply. The thread shall be in sizes AA, B and E. The color of the thread shall be black.

3.3.6.2 Thread, sole stitching. The thread for sole stitching shall be polyester and shall conform to type I, class 1, subclass C of V-T-285. The color of the running thread shall be black and the color of the shuttle thread shall be natural. The requirements for colorfastness shall not apply. The running thread shall be size 10, 3 ply. The shuttle thread shall be size 10, 3 ply or size 10, 6 ply.

3.3.6.3 Thread, inseaming. The inseaming thread shall be polyester, size 10, 3 or 6 ply conforming to type I, class 1, subclass C of V-T-285. The color shall be natural.

3.3.7 Bottom filler. The bottom filler shall be either thermoplastic or cold process type.

3.3.7.1 Thermoplastic bottom filler. The thermoplastic type shall consist of a mixture of ground cork and a suitable thermoplastic binder in proportion of a minimum of 2-3/4 parts by volume of cork to each one part of binder. The cork granules shall be free from bark. The ground cork and the binder shall be thoroughly and evenly mixed. The binder shall be water insoluble. The binder shall have a softening point of at least 125°F and a maximum penetration of 85 mm with a 200 gram load for 60 seconds at 77°F when tested as specified in 4.4.1.

3.3.7.2 Cold process bottom filler. The cold process type shall be spreadable without the use of heat. It shall consist of a mixture of ground cork and a suitable binder in the proportion of two parts cork to one part binder by volume. The cork granules shall be free from bark. When spread filler is dry and set, it shall consist of four parts cork to one part binder. The binder shall be water insoluble and shall have a softening point of at least 150°F when tested as specified in 4.4.1. The binder shall be tested after evaporation to a constant weight level. As an alternate, a cold process bottom filler applied by semiautomatic equipment or method may be used. The alternate filler shall consist by weight of a mixture of one part ground cork to five parts suitable binder. Upon loss of solvent, the alternate filler shall consist by weight of one part cork to 3-3/4 parts binder. The binder shall be water-insoluble and have a softening point of at least 125°F when tested as specified in 4.4.1.

3.3.8 Wood shank. The shank shall be made from a straight grain birch wood or from equivalent straight grain hardwood. The width of the shank shall be 1-1/16 + 1/16 inches. The thickness of the shank shall be 0.125 to 0.156 inch. The edges of the shank shall be beveled.

3.3.8.1 Shank fitting sizes. The wood shanks shall be carefully selected for correct size in accordance with commercial practice. The shanks shall be inserted in position, filling the cavity from the ball line rearward to slightly behind the heel breast line. The forward end of the shank shall be flush with the insole, fit the contour of the shoe bottom back of the ball line, and be attached with tar or an adhesive specified in 3.3.10 that is compatible with the wood.

* 3.3.9 Laces. The laces shall conform to type II, class 1 or class 4 of V-L-61. The color of the laces shall be black. The length of the laces shall be 26 inches for shoe sizes 4 to 12, and 27-1/2 inches for shoe sizes 12-1/2 to 14.

3.3.10 Adhesive. The adhesives used for bonding the various parts of shoes shall be of the following types:

(a) Natural rubber latex.
(b) Synthetic rubber latex (including chloroprene).
(c) Natural rubber solvent cement.
(d) Synthetic resin cements.
(e) Synthetic rubber solvent cement.
(f) Synthetic cement for sole stock fitting.

3.3.11 Wax, inseam sewing and sole stitching. The wax shall be white or golden and shall be a permanently plasticized resin. The wax, in the temperature range of normal machine use, shall thoroughly wax the threads described in 3.3.6.2 and 3.3.6.3.

3.3.12 Repairers. Repairers shall be liquid spray, crayon, or paste-type applied by hand. The color shall match the upper leather and shall have sufficient coverage to correct minor surface imperfections of the leather.

3.3.13 Renovators. Renovators, used in lieu of or in addition to repairers, shall match the color of the upper leather. Application may be by sponge or spray method.

3.3.14 Fillers. Fillers, where used, shall be capable of providing a foundation for the application of top finish and shall be applied by sponge or spray method.

3.3.15 Top finish. The top finish shall be applied by sponge or spray method and shall be a material compatible with, and capable of wetting into and binding the previously applied coats. The finish shall match the dyed leather and shall provide luster.

* 3.4 Design. The design shall be for a full welt around the heel construction blucher shoe cut from grain-out leather. The shoe shall have a full plain toe vamp, oxford blucher quarter, leather quarter lining, and leather vamp lining. The shoe shall have no metal parts except for the safety steel box toe and side lasting staples. The shoe shall have a raw edge around the top of the quarter (see 3.5). The shoe shall have a leather or Texorist insole, rubber midsole, and a traction tread sole and heel with chevron design.

* 3.4.1 Safety toe performance requirements. The finished footwear shall conform to the class 75 performance requirements of ANSI Z41.1. The label requirements shall be in accordance with 3.7.4.4.

3.5 Patterns and dies. A standard set of paper patterns and their markers will be furnished by the Government as a basis from which the contractor's dies or patterns shall be made. The contractor shall furnish the wood insole rounding patterns conforming to paper patterns furnished by the contracting activity. The patterns, markers, and contractor's cutting dies and patterns shall consist of the component parts specified in table VII. Modifications shall be made to the patterns at the top edge and front edge of the quarters to elimate the allowance for the folded edge and to produce a raw edge on the finished shoe. The Govenment patterns and markers shall not be altered in any way and shall be followed except for top and front of the quarters, and for lasting allowance which shall be determined by the contractor. The long side of the leather quarter lining may be used with an inverted "V" removed in the shoe back center. The height of the cut for this inverted "V" shall be a maximum of 1-1/4 inches from the bottom of the lasting allowance on a size 4 and grading upward to a maximum height of 1-1/2 inches on size 14.

TABLE VII. Contractor's dies and patterns

Component parts	
Vamp	Whole and half sizes
Quarter inside	Whole and half sizes
Quarter outside	Whole and half sizes
Tongue 1/	8 sizes
Vamp lining 1/	8 sizes
Quarter lining inside	Whole and half sizes
Quarter lining outside	Whole and half sizes
Insole rounding	Whole and half sizes
Markers	
Vamp 1/	8 sizes

1/ XN has 7 sizes only.

3.6 Lasts. The shoe shall be made on standard U.S. MIL-7 (MIL-L-43585) men's safety toe lasts, all sizes and widths of which will be loaned to the contractor by the Government.

3.7 Construction.

3.7.1 Cutting uppers. The uppers shall be cut from grain-out leather. The leather shall conform to the thickness specified in table VIII. Vamps shall be cut from the bend area of the side leather.

TABLE VIII. Thickness of upper leather

Part 1/	Minimum (ounces)	Maximum (ounces)
Vamp	4	3/4
Quarter	3-1/2	4-3/4
Tongue	3	3-1/2

1/ Except for tongue, none of these parts shall be split or shaved. No parts cut off stretch shall be accepted.

3.7.2 Skiving. The skiving of upper leather parts shall conform to the requirements specified in table IX.

.BLE IX. Skiving requirements

Part	Size	Location	4-3/4 oz leather	ather scarf width inches oz ather	3-1/2 oz leather	3 oz leather	2-1/2 oz leather	Scarf edge thickness (millimeters) ± 0.1 mm
Quarters	Flesh	Vamp line	3/16 ± 1/32	-	3/32 ± 1/64	-	-	0.8
Vamps	Flesh	Throat and wings	1/4 ± 1/16	/64 ± 32	-	-	-	0.3
Tongues	Flesh	Bottom edge				11/64 ± 3/64	-	0.3
Quarter lining	Flesh	Long and short piece vamp			3/8 ± 1/8	-	3/64 ± 1/64	0.75
Quarter lining	Flesh	Long piece closing edge			3/8 ± 1/8	-	3/64 ± 1/64	0.8

NOTE: A dash indicates that the scarf width i e same as indicated in the preceding column.

3.7.3 Crimping. The vamps may be crimped, if necessary, to insure tight lasting.

3.7.4 Marking.

3.7.4.1 Identification. The inside quarter of each shoe shall be marked on the grain side with the correct size and width, the contractor's identification symbol, and the month and year (expressed numerically) of the date of contract. This marking shall be placed approximately in the center of the inside quarter, 1/2 ± 1/8 inch below the top. The outside quarter of each shoe shall be marked on the grain side with the letters EH. The markings shall be impressed into grain side of the leather in such a manner as to be permanently visible without cutting through at any point. Figures shall be Arabic-type and letters Gothic. The figures and letters shall be a minimum of 9/32 inch and a maximum of 3/8 inch in height except that the letters EH (electrical hazard) shall be a minimum of 1/2 inch and a maximum of 9/16 inch in height. The contractor's symbol shall be in a block as shown by the following example: 10W /AB/ 9-76.

3.7.4.2 Description. Each heel pad shall contain a description marking imprinted with an electrical embossing machine using indelible ink. The letters shall be Gothic type and shall be a minimum 3/16 inch in height. The legend shall read as follows:

THESE SHOES ARE ELECTRICAL HAZARD PROTECTIVE WITH STEEL TOES

3.7.4.3 Warning tag. A warning tag shall be furnished by the contractor and shall be attached to each pair of shoes. The tag shall be of cardboard and white or light in shade to permit easy reading of the printed marking. The printed marking shall be as follows:

WARNING

These electrical hazard shoes are not recommended for use while working in the vicinity of live circuits exceeding 600 volts. The shoes should be tested periodically to ensure that current leakage does not exceed 14 milliamperes when 14,000 AC volts (RMS Value) 60 Hz (cycles) are applied for 1 minute. For use on circuits exceeding 600 volts, consult your safety officer. The shoes should not be worn in flooded areas where water might saturate the shoe above the welt line.

* 3.7.4.4. Ansi labeling. The shoes shall be labeled to indicate that they conform to class 75 safety toe requirements specified in ANSI Z41.1. Each shoe shall have an indelible cloth label sewn to the inside of the shoe on the tongue lining. The numbers and letters shall be a minimum of 3/16 inch in height. The label inscription shall be as follows:

ANSI

Z41.1 - 1967/75

3.7.5 Upper fitting. The upper fitting stitch types, thread sizes and stitches per inch shall conform to the requirements specified in table X. The stitch types specified in table X shall conform to FED-STD-751 (see 3.7.22).

TABLE X. Upper fitting stitching requirements

	Stitch type	Number of stitches per inch	Thread size	
			Needle	Bobbin or looper
Quarter lining closing	301 or	8-10	B	B
	401		B	AA
Quarter lining "V" notch (when used)	301 or	8-10	B	B
	401	8-10	B	AA
Top stitching	301	8-10	B	B
Tongue attaching	301	8-10	E	E
Backseam closing	401	11 minimum	E	B
Vamping	301	8-10	B	B
Barring	301	8-10	B	B

3.7.5.1 Backseam stitching (quarters). The backseam shall be closed with one row of stitching closely positioned to the edge of the quarters, and rubbed down.

3.7.5.2 Quarter lining and top stitching. The quarter linings shall be lapped and stitched with two rows of stitching spaced 1/8 ± 1/32 inch apart. If inverted "V" seam lining is used, the "V" seam shall be closed upwards from the lasting edge. The closing shall be done with one row of stitching. This seam shall extend 1/16 to 1/8 inch on the flesh side and the seam shall be rubbed down. The assembled linings shall be top stitched to the quarters with a single row of stitching using a round point needle which shall pass through the quarter lining and catch the dog ear even to the top of the inside quarter. Full advantage of the curves on the top stitching machine must be taken since the pattern is form fitting and will not allow use of a flat bed under trimming machine. The quarter lining shall be sufficiently undertrimmed below the quarter top raw edge of the quarters to minimize exposure of the lining.

3.7.5.3 Tongue fitting. The tongue shall be placed between the vamp and vamp lining and stitched with two rows of stitching, spaced 1/8 ± 1/32 inch in from center to throat.

3.7.5.3.1 Lacing for lasting. The machine lacing for lasting shall provide a 9/16 ± 1/16 inch opening after lasting. The quarters shall be laced in the three lower pairs of lacing holes.

3.7.5.4 Lace holes. On each quarter for lacing the shoe, there shall be five punched holes on all sizes. Eyelets shall not be permitted. The holes shall be of a diameter to adequately accomodate a round shoe lace not greater than 1/8 inch diameter. The edge of the holes shall be not less than 1/4 inch or more than 5/16 inch from the edge of the quarter. The holes shall be spaced evenly from blucher nose to top of quarter. The holes shall be punched cleanly and neatly.

3.7.5.5 Vamping, barring. The vamping shall be done with a single row of stitching close to the edge of the quarter and a single row 1/4 +1/16, -1/32 inch from the first row. Two rows of barring, 3/8 by 1/8 ± 1/16 inch shall be included inside the vamping rows. Barring may be accomplished not more than 1/8 inch above the inside vamping row to provide for one continuous vamping and barring on single needle machine, or by the automatic method with tying or locking ends.

3.7.6 Rubber midsole and outsole stock fitting. The inner surfaces of the rubber outsole and rubber midsole shall be roughed their full length with a coarse abrasive paper prior to cementing. The entire roughed surfaces of the midsole and outsole shall be coated with a cement specified in 3.3.10 that is compatible with rubber compounds. The combined midsole-outsole shall have a minimum bond strength of 12 pounds average per linear inch of width with no individual value below 10 pounds when tested as specified in 4.4.2.4.

3.7.7 Insoles. Insoles shall be sorted for varying fibers to get uniform channeling and cased for even weight. The insoles shall be fleshed. The insoles shall be rounded to pattern. Die cut insoles will be accepted provided they conform to insole patterns loaned by the Government. A ± 1/64 inch tolerance from the Government loaned insole patterns will be allowed.

3.7.7.1 Insole channeling. The insole, except the stuck-on rib insole, shall be channeled including heel seat to provide for a welted heel seat. The insoles may be chamfered. The channel leaf and outside lip shall be not less than one third the thickness of the insole, and the channel leaf shall be slightly thicker than the lip. The in-between substance shall be 1/64 to 1/32 inch greater than the combined thickness of the leaf and lip. The channel margin shall be a minimum of 5/32 inch and the maximum shall not exceed the required extension on the finished shoe (see 3.7.16). The lips shall be cemented with a latex adhesive specified in 3.3.10 (a) or (b) and pressed together at lip setting to incline inward and be a minimum of 1/8 inch in height.

* 5.1.2 Level C. Footwear covers shall be packaged to afford adeq

3.7.7.1.1 Assembly of stuck-on rib. When assembled, the stuck-on rib insoles shall meet the requirements of 3.3.5.3.

3.7.7.2 Insole backing. All insoles except stuck-on rib insoles (see 3.3.5.3.1) shall be reinforced by the application of the cotton duck specified in 3.3.5.1. The duck shall be coated on one side with any adhesive compound specified in 3.3.10 and shall be bonded to the flesh side of the insole. The duck shall cover the entire area between the ribs, extend to the top of the upstanding rib, be firmly and smoothly fitted and bonded to the base and the lip, and be trimmed to the required rib height. The insole shall be cemented with any adhesive as specified in 3.3.10 on the flesh side prior to attaching the coated duck.

3.7.8 Lasting. Uppers may be conditioned by any suitable means except dipping in water. The correct size and width of uppers, lasts, and insoles shall be assembled. Insoles shall be tacked to the last with not less than five tacks: one of the five tacks shall be in the center of heel seat, one at shank, one at each side of the ball, and one at the toe. Staples in lieu of tacks may be used for attaching the insole to the last. Edges of the insole shall be flush with the last bottom at all points. The heel seat of the insole shall be smooth and even with the heel seat of the last. Counters shall be assembled in accordance with sizing schedule so that they will fill the counterpockets. The counters shall be molded to insure that they are caught by the inseam stitching around the periphery of the heel seat. Counters shall be cemented on both sides, except that cementing of polyethylene counters is optional. Uppers, of the correct size and width, and counters shall be assembled to the lasts. Uppers shall be drawn over the lasts with tension on the pulling-over machine to assure that the quarters at the blucher points and the vamps are down to the last, and that the blucher noses are even. Sides of shoes shall be spindled, drawing the upper snugly to the last, and then side-lasted so that when stapled, the upper will be held securely to the last. Spacing between staples shall be 3/4 $\pm$ 1/4 inch. The heel seat shall be wiped-in and secured to provide for a welt around the heel seat.

3.7.8.1 Steel toe lasting , The vamp shall be laid back sufficiently to accommodate the steel toe. The correct size steel box toe shall be inserted. The felt strip described in 3.3.5.2.1 shall be superimposed on the vamp lining directly to the rear and abutting the edge of the steel box toe. Then the masking tape described in 3.3.5.2.2 shall be placed to completely cover the curved felt and mask the joint of the felt strip and the steel box toe edge. The polyurethane cushion pad described in 3.3.5.2.3 shall be placed so as to cover the steel box toe, felt strip, and masking tape. Then the vamp shall be lasted back in place, the toe shall be firmly and smoothly wiped in, and the toe wire or nylon filament shall be attached securely around toe at the base of the insole rib providing a shoulder for subsequent inseaming. As an alternate, any suitable equipment or method may be used to smoothly and securely attach the toe around the base of the insole rib.

3.7.8.2 Time allowance on lasts. The shoes shall remain on the lasts a sufficient length of time to thoroughly dry the counters, uppers, and insoles.

3.7.9 Inseaming. Inseaming shall be done by machine, with the thread waxed and with tension on the thread, using not larger than a number 41 needle and not less than 3-1/4 stitches to the inch. The welting shall be inseam stitched to the bottom of the insole rib completely around the periphery of the shoe to provide for a welted heel seat. As an alternate, the operation for side-lasting and inseaming may be performed simultaneously with any suitable equipment or method. Prior to inseaming, the sides of the shoe shall be spindled, drawing the upper snugly to the last.

3.7.10 Tack and staple pulling, inseam trimming. All tacks, staples, and toe wire shall be removed from the insole bottom and no broken tack or staple points shall remain. The inseam shall be closely trimmed, without cutting or damaging stitches. The welt shall be laid out smoothly to lie flat around the periphery of the shoe.

3.7.11 Shank fitting, bottom filling. The shanks shall be selected for correct size as specified in 3.3.8.1. The bottom filler shall be firmly pressed into the insole channel from the toe to the forward end of the shank with a uniform and even surface. The bottom shall present a flat, smooth surface for sole laying. The shank and heel cavities shall be filled with bottom filler flush with the welting.

3.7.12 Sole laying. The shoe bottoms shall be thoroughly cemented with any adhesive specified in 3.3.10 and after the combined rubber midsole and rubber outsole have been cemented as specified in 3.7.6, the midsole shall be cemented over the entire inner surface and shall be laid on a sole laying machine with pressure. The soles shall be of adequate size and shall be laid evenly in order to prevent the sole stitching from running into the plaque area and to allow for the specified edge extension.

3.7.13 Rough rounding. Soles shall be uniformly rounded on a rough-rounding machine for the specified edge extension of sole and welt as specified (see 3.7.16).

3.7.14 Sole stitching. Soles and welts shall be stitched together on a lockstitch machine using waxed polyester thread size 10, 3 ply for the running thread and size 10, 3 ply or size 10, 6 ply for the shuttle thread with a No. 45 round blade chisel point awl and not larger than a No. 45 needle, using 5-1/2 to 7 stitches per inch except that there shall be no more than 3-1/2 stitches per any one-half inch length in the ball and toe sole stitching. The stitching shall be laid on the surface of the welt and close to the outer edge of the welt on the finished shoe. The stitching shall be with the lock (peg) just under the surface of the outsole.

3.7.15 Heeling . Heels as specified in 3.3.3.2 shall be of correct size and shall be attached without the use of heel nails. The heel cup and sole heel seat shall be roughed with a wire brush. One coat of any adhesive specified in 3.3.10 that is compatible with rubber compounds shall be applied to the roughed area of the heel cup and sole heel seat, allowing sufficient drying time to insure the required bond. As an alternate, the drying time may be accelerated by the use of a portable dryer. The heels shall then be spanked on using a heeling machine, or any other suitable equipment. The bond strength, measured in peel shall be a minimum of 25 pounds per inch when tested as specified in 4.4.4.

3.7.16 Edge trimming. Edges shall be trimmed square. The finished minimum extension of outside ball shall be 3/16 inch, and of toe and inside ball shall be 1/8 inch, when measured at right angles from the upper.

3.7.17 Heel finishing. The heel shall be trimmed and smoothly scoured. The beveled breastline shall not be scoured.

3.7.18 Finishing.

3.7.18.1 Preparation. The shoes shall be cleaned by removing accumulated dirt, such as wax or cement, and the surface of the leather shall be conditioned to receive further application of the finish. All thread ends shall be trimmed.

3.7.18.2 Treeing. The shoes shall be treed while on the lasts or on a treeing machine, using right and left tree feet conforming to the MIL-7 safety toe last. All wrinkles shall be removed. No material shall be used that may injure the leather or thread.

3.7.18.3 Final finish. Shoes shall be repaired and filled. Materials and methods shall be as specified in 3.3.12 to 3.3.15 inclusive.

3.7.19 Tacks and staples. Tacks and staples that have been left protruding through or around the insole shall be pulled out. A mechanical tack detector or other suitable method may be used to indicate the presence of any protruding tacks or staples inside the shoe.

3.7.20 Heel pads. The flesh side of the leather heel pad shall be coated with any adhesive as specified in 3.3.10. Heel pads shall be pressed firmly and completely into the heel seat. Heel pads shall be fitted in accordance with size schedule.

3.7.21 Lacing. The shoes shall be mated. A lace shall be inserted through the top lacing hole of the outside quarter of each shoe, and the two laces shall be tied firmly together.

3.7.22 Repairs of stitching.

3.7.22.1 Type 301 and 401 upper fitting stitches. Repairs of type 301 and 401 stitches shall be as follows:

a. When thread breaks or bobbin run-outs occur during stitching, the stitching shall be repaired by restarting the stitching one stitch in back of the end of stitching.

b. Thread breaks, or two or more consecutive skipped or run-off stitches, noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start one stitch in back of the defective area and continue one stitch beyond the defective area onto the existing stitching.

3.7.22.2 Inseaming and sole stitching. Repairs of inseaming and outsole stitches shall be in accordance witb 3.7.22.1 a and b above and in addition, thread cuts occurring as a result of trimming operations shall be repaired in the same manner as specified for thread breaks.

3.8 Replacement of defective components. During the cutting and manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.1, 4.4.2.2 and 4.4.3 shall be removed from production and replaced with non-defective and properly matched components.

3.9 Electrical test of finished shoe. Each finished shoe shall be capable of withstanding a proof test of an alternating current potential of 10,000 volts (rms) 60 Hz (cycles) for 3 minutes without a leakage current of more than 10 milliamperes when tested as specified in 4.4.5. In addition, each finished shoe shall be capable of withstanding the application of 14,000 volts (rms) 60 Hz (cycles) break down voltage test for 1 minute without showing a leakage current in excess of 14 milliamperes when tested as specified in 4.4.5.

3.10 Workmanship. The finished shoes shall conform to the quality and grade of product established by this document. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

* 4.3 First article inspection. When a first article is required (see 6.2) it shall be examined for the defects specified in 4.4.3 and tested as specified in 4.5.4, 4.5.5 and 4.5.6. The presence of any defect or failure of any test shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document. In addition, testing shall be performed on components and materials listed in table XI. When the data in the "Number of determinations per sample unit" and "Results reported as" columns are not specified in table XI, they shall be as required by the referenced test methods. The upper leather sampling procedure for determining stitch tearing strength shall be in accordance with FED-STD-311 and the lot shall be rejected if more than three specimens have stitch tearing strength values of less than 30 pounds. The vamp and quarter lining leather sampling procedure shall be in accordance with FED-STD-311. All test reports shall contain the individual values utilized in expressing the final result. The lot shall be unacceptable if one or more sample units, the lot average, or the composite sample fails to meet any requirement specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE XI. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement para.	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Upper leather (1 side)	Stitch tearing strength	3.3.1.1	2151 1/	X	-	-	-	-	See 4.4.1
Insole (Texorist)	Copper-8-quinolinolate	3.3.1.2	2/	-	-	-	-	-	
Vamp and quarter lining	Tannage	3.3.1.3	2/	-	-	-	-	-	
	Material identification	3.3.1.3	2/	-	-	-	-	-	
	Thickness	3.3.1.3	1011 1/	X	-	-	-	-	See 4.4.1
	Paranitrophenol	3.3.1.3	6711 1/	-	3/	-	-	-	
Heel pads (1 pad)	Tannage	3.3.1.4	2/	-	-	-	-	-	
	Material identification	3.3.1.4	2/	-	-	-	-	-	5 heel pads
	Thickness	3.3.1.4	1011 1/	X	-	-	-	-	
	Paranitrophenol	3.3.1.4	6711 1/	-	3/	-	-	-	
Counters, polyethylene (1 pair)	Density	3.3.2.2	5011 4/	X	-	-	-	-	
	Melt index	3.3.2.2	D 1238 5/	-	X	-	-	-	8 counters
	Tensile strength	3.3.2.2	2/	-	-	-	-	-	
	Elongation	3.3.2.2	2/	-	-	-	-	-	
Midsoles (1 pair)	Non-marking	3.3.3.3	4.5.1	X	-	-	X	-	
	Hardness (before and after aging)	3.3.3.3	D 2240 5/ 6/	X	-	-	X	-	2 pairs

* TABLE XI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement para	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Midsoles (1 pair) (cont'd)	Cut growth	3.3.3.3	4.5.2	X	-	2	-	Percent	
	Thickness	3.3.3.3	2/	-	-	-	-	-	
Vinyl welting (1 yard)	Material identification	3.3.3.4	2/	-	-	-	-	-	
	Hardness (Shore A durometer)	3.3.3.4	D 2240 2/ 5/	-	-	-	-	-	
	Specific gravity	3.3.3.4	2/	-	-	-	-	-	
	Brittle point (before and after aging)	3.3.3.4	D 746 2 5/	-	-	-	-	-	
	Tensile strength	3.3.3.4	D 412 (Method A - Die E) 2/ 5/	-	-	-	-	-	
	Elongation	3.3.3.4	D 412 (Method A - Die E) 2/ 5/	-	-	-	-	-	
	Water absorption	3.3.3.4	2/	-	-	-	-	-	-
Steel box toe (1 pair)	Material identificaton	3.3.4.1	2/	-	-	-	-	-	-
	Hardness	3.3.4.1	E 18 5/	X	-	3	-	Hardness number	3 box toes
	Carbon content	3.3.4.1	2/	-	-	-	-	-	-
	Thickness	3.3.4.1	2/	-	-	-	-	-	-
	Coating or plating identificaton	3.3.4.1	2/	-	-	-	-	-	-

TABLE XI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement para	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Results reported as: Sample unit
Masking tape	Material identification	3.3.5.2.2	2/	-	-	-	-	-	-
Toe cushion (polyurethane)	Material identification	3.3.5.2.3	2/	-	-	-	-	-	-
Insole ribbing fabric (1 yard full width)	Material identification	3.3.5.3.1	2/	-	-	-	-	-	-
	Weight	3.3.5.3.1	5041 7/		-	-	--	-	-
	Yarns per inch	3.3.5.3.1	5050 2/ 7/	-	-	-	-	-	-
	Breaking strength	3.3.5.3.1	5100 2/ 7/	-	-	-	-	-	-
Bottom filler (1 pound)	Material identification	3.3.7	2/	-	-	-	-	-	-
	Proportion of cork to binder	3.3.7.1 and 3.3.7.2	2/	-	-	-	-	-	-
	Solubility of binder in water	3.3.7.1 and 3.3.7.2	2/	-	-	-	-	-	-

* TABLE XI. Component tests (cont'd)

Component and lot size expressed in terms of	Characteristic	Requirement para	Test method	Requirement applicable to: Sample unit	Requirement applicable to: Lot average	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Bottom filler (1 pound) (cont'd)	Softening point of binder	3.3.7.1 and 3.3.7.2	E 28 5/	-	X	2	-	Avg. of 2 determinations to nearest 1°F	1 pound composite
	Penetration of binder	3.3.7.1	D 5 5/	-	X	3	-	Avg. of 3 determinations to nearest mm	
Wood shank	Material identification	3.3.8	2/	-	-	-	-	-	-
	Width and thickness	3.3.8	2/	-	-	-	-	-	-
Adhesive	Material identification	3.3.10	2/	-	-	-	-	-	-
Wax	Material identification	3.3.11	2/	-	-	-	-	-	-

1/ Refers to FED-STD-311.

2/ A certificate of compliance shall be submitted and will be acceptable for the stated requirement.

3/ Requirement applicable to composite.

4/ Refers to FED-STD-406.

5/ Refers to ASTM standard.

6/ Aging shall be at a temperature of 158 ± 1°F for 166 ± 1 hour in accordance with ASTM D 573 procedures for accelerated aging.

7/ Refers to FED-STD-191.

4.4.1.1 Examination of Texorist insoles (when used). The Texorist insoles shall be examined for the defects listed below. The sample unit shall be one insole. The lot size shall be expressed in units of one insole. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

Any hole, cut, tear, or gouge.
Any brittle area or evidence of delamination.
Thickness not as specified.
Sole outline does not conform to required pattern.
Not cut in specified direction.

4.4.2 In-process inspection. Inspection shall be made at any point or during any phase of manufacturing to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance.

4.4.2.1 In-process examination of uppers after all fitting. The upper assembly shall be examined for the defects listed below. The lot size shall be expressed in units of one upper. The sample unit shall be one completely fabricated upper assembly prepared for lasting. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

Examine	Defect	Classification Major		Minor
Construction and workmanship	Any component missing or not specified type.	X		
	Any component misplaced or not affixed as specified.		*	
	Vamp throat or wings not skived.	X		
	Quarter not skived as specified.	X		
	Backseam not rubbed down.			X
Quality of leather	Thickness more than 1/2 ounce less than the minimum specified.	X		
	Thickness less than specified minimum but not exceeding 1/2 ounce less.			X
	Thickness more than maximum specified.			X
	Rough fiber on flesh side of quarters.		*	
	Off-stretch cut.	X		
	Slaughter cut.		*	
Lining leather (vamp)	Not soft and pliable.	X		
Stitching of quarter lining "V" notch when used	Quarter lining "V" notch has less than 8 or more than 10 stitches per inch.			X

* 4.4.2.2 In-process examination of shoe before bottom filling. The partially fabricated shoe shall be examined for the defects listed below. The lot size shall be expressed in units of one shoe. The sample unit shall be one partially constructed shoe at a point after lasting and attachment of shank but before the application of the bottom filler. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects.

NOTE: Defects designated by an asterisk shall be scored as major when seriously affecting serviceability and minor when affecting serviceability but not seriously.

Examine	Defect	Classification Major		Minor
Bottom of shoe	Any component missing or not specified type.	X		
	Shank not properly positioned or not fitting contour of shoe bottom.		*	
	Shank wrong size, malformed, or not securely attached.			X
	Any tear in cotton duck in stitch area:			
	-more than 3/4 inch	X		
	-more than 1/2 inch but not more than 3/4 inch			X
	Any insole tack, staple, or anchor tack not removed.	X		
	Upper damaged.		*	
	Poor heel seat, side, or toe lasting.		*	
	Spacing of staples not as specified.			X
	Inseam not properly trimmed.		*	
	Less than 3 stitches per inch on inseam.	X		
	Less than 3-1/4 but not less than 3 stitches per inch on inseam.			X
	Inseam stitches broken, two or more consecutive skipped inseam stitches, or inseam stitches cut or damaged during trimming operation:			
	-not repaired.	X		
	-not repaired as specified.			X
	Inseam stitches not at bottom of insole rib or not in welt groove.		*	
	Broken insole rib.	X		
	Counter not caught in inseam stitching.	X		
	Welt not properly laid out, i.e., not flat.			X
Upper part of shoe	Upper not tightly pulled down to last.			X
	Lace opening less than 1/2 inch or more than 5/8 inch			X
	Quarter not laced as specified.			X
	Steel box toe out of alignment or crooked.		*	
	Steel toe missing.	1/		

1/ Any steel toe found missing in the sample shall be classified as a critical defect and result in rejection of the lot represented.

4.4.2.3 In-process testing of stuck-on rib insole. When used, the stuck-on rib shall be tested for compliance with the requirements of 3.3.5.3.2. Method 2061 of FED-STD-311 shall be used for shear strength test. Method 2171 shall be used for stitch tear strength. The sample unit shall be two ribbed insoles, one for each test, and the sample size shall be 5, regardless of the lot size. Requirements are applicable to the sample unit for individual unit requirements and lot average for average requirements. The lot shall be unacceptable if one or more sample units or the lot average of all sample units fail to meet the requirements specified. All test reports shall contain the individual values utilized in expressing the final results.

4.4.2.4 In-process testing of outsole and rubber midsole assembly. The rubber outsole and midsole assembly shall be tested to determine compliance with bond strength requirement of 3.7.6. The test method shall be as specified in 4.5.3. The sample unit shall be one cemented midsole and outsole unit. The lot size shall be expressed in units of one cemented midsole and outsole unit. All test reports shall contain the individual values utilized in expressing the final result. The requirement is applicable to the sample unit. The lot shall be unacceptable if one or more sample units fail to meet the requirement specified. The sample size shall be as follows:

Lot size	Sample size
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

4.4.3 End item visual examination. The shoes shall be examined for the defects listed below. The lot size shall be expressed in units of one shoe. The sample unit shall be one completely fabricated shoe and the selection shall be by pairs. The inspection level shall be II and the AOL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total (major and minor combined) defects. Defects of pairing shall be classified as a single defect. The vamp, and inside and outside quarters of each shoe shall be examined for break of leather in accordance with the procedure outlined below:

(a) To examine the vamp, the shoe shall be held in an upright position with both hands. The toe of the shoe shall face away from the examiner. Position thumbs on top of the vamp approximately half way between box toe line and blucher noses and 1 inch to 1-1/2 inches apart. Press downward with thumbs so as to form grain surface into a concave surface and observe the break pattern. Any vamp exhibiting a break pattern at any location comparable to a break pattern represented by numbers 5 through 8 on the Satra scale (see 6.5) shall be scored as a defect. The defect shall be scored regardless of the direction in which the break pattern appears.

(b) To examine the quarters, the index finger and the second finger held in a "V" position shall be placed against the inside of the quarter in the midsection area between the blucher nose and counter area. The thumb shall be placed on the outside of the quarter directly above the slotted fingers. Pressure shall be applied to the quarter with the thumb and fingers so as to impart a break pattern to the leather. The break pattern of the quarter shall be observed and compared with the break patterns of the Satra scale. Any quarter exhibiting a break pattern comparable to a break pattern represented by numbers 5 through 8 on the Satra scale shall be scored as a defect. The following note, regarding defects visible at a distance of 3 feet, is not applicable to examination of the break pattern.

NOTE: Appearance defects shall be scored only when the condition is plainly visible at a distance of 3 feet or more. Defects designated by an asterisk (*) shall be scored as major when seriously affecting serviceability or appearance, and minor whan affecting serviceability or appearance but not seriously.

Examine	Defect	Classification Major		Minor
Pairing	Not properly mated, i.e., not right and left of same size.	X		
	Variation in color or appearance.		*	
Cleanness	Any non-removable spot, stain, or foreign matter affecting appearance.			X
Color and finish	Not specified color.	X		
	Finish streaky, chipped, or flaky on upper.			X
Design, type, and size	Not as specified.	X		
Upper leather	Leather deeply snuffed, i.e., fiber structure damaged.	X		
	Any grub or tick damage, scratch, brand, or boney leather.		*	
	Fat wrinkles or veins.		*	
	Stretchy vamp.	X		
	Leather embossed or printed.	X		

Examine	Defect	Classification Major	Minor
Break pattern of vamps and quarters	Break pattern comparable to break pattern represented by numbers 5 through 8 on Satra break scale.	X	
NOTE:	Leather exhibiting a break pattern comparable to the numbers indicated above is considered to be loose, pipey, or flanky. The break shall be determined in accordance with 4.4.3.		
Lining leather (quarter)	Quarter lining not soft and pliable.	X	
	Slaughter cut, seriously affecting serviceability.	X	
	Pipey or flanky.	X	
Construction and workmanship	Any cut, tear, hole, repair, or factory damage.		X
	Any component or assembly omitted or misplaced, operation omitted or not properly performed (unless otherwise specified herein).	*	
	Loose leather fibers exposed along cut edge of quarter vamp line.	X	
Seams and stitching (upper)	Open seam not repaired.	X	
	Repair of open seam not as specified.		X
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken, or when two or more consecutive skipped or run-off stitches occur. On multiple stitched seams, a seam is considered open when either one or both sides of the seam are open. When any of the above conditions occur on the inside of the shoe, it shall be scored as an asterisk (*) defect.		
	Loose tension resulting in a loosely secured seam.	*	
	Tight tension resulting in puckering or cutting of leather.	*	

Examine	Defect	Classification Major	Minor
Seams and stitching (upper) (cont'd)	Wrong stitch or seam type.	X	
	Any row of stitching with less than 8 stitches per inch but not less than 6 stitches per inch.		X
	Any row of stitching with less than 6 stitches per inch.	X	
	More than the specified maximum number of stitches.	*	
	NOTE: A plus tolerance of 3 stitches per inch will be allowed when stitching over heavy places or turning sharp corners.		
	Gage of stitching not as specified or irregular.	*	
	Stitching omitted where required.	*	
	Needle holes or needle chew.	*	
	Thread ends not trimmed throughout shoe.		X
Outsole stitching	Less than 4-1/2 stitches per inch.	X	
	Less than 5-1/2 stitches per inch but not less than 4-1/2 stitches per inch.		X
	More than 7 stitches per inch except in ball and toe sole stitching.		X
	NOTE: More than 3-1/2 stitches per any 1/2-inch length in the ball and toe sole stitching shall be classified as a minor defect. The ball and toe sole stitching shall be defined as that portion of the sole stitching that runs from the inside ball and around the toe to the outside ball.		
	Broken stitch or two or more consecutive skipped stitches:		
	-not repaired.	X	
	-not repaired as specified.		X
	Sole stitching not properly gaged.	*	
	Any stitching not visible on surface of outsole.	*	
	Stitch locks laying on surface.	*	
	Stitches running into traction tread plaque.	*	
	Outsole or welt damaged by stitches.	X	
Counters	Rolled or curled counter.	*	
	Soft counter.		X

examination. The shoes shall be examined for the defects 4.4.1 and item visual e

Examine	Defect	Classification Major	Minor
Edge making	Edge trimmed into sole stitching.	*	
	Edge not trimmed or irregular affecting appearance.		X
	Sole extension less than the specified minimum.	*	
	Checked sole, i.e., separation between outsole and midsole.	*	
Heel finishing	Heel not finished square, i.e., has pronounced flare or taper.		X
	Heel not finished smooth affecting appearance.		X
	Checked heel, i.e., separation at heel and outsole.	*	
	Crooked or wrong size heel.		X
Lacing holes	Number of lacing holes not as specified but each row has the same number.		X
	Not the same number of lacing holes in each row.	X	
	Lacing holes not properly spaced within row or misalignment between the rows to an extent interfering with proper lacing.	*	
	Edge of any lacing hole less than 3/16 inch from the edge of the quarter.	X	
	Edge of any lacing hole less than 1/4 inch but not less than 3/16 or more than 5/16 inch from the edge of the quarter.		X
Inseaming	Grinning seam, i.e., thread exposed.	X	
	Strained seam, i.e., needle holes visible but thread not exposed.		X
	Dropped welting, skipped stitches, or cut inseam.	X	
Lining	Vamp lining having excessive fullness, loose, or wrinkled.	*	
Insole	Short or long.	*	
	Any protruding point of lasting staple, or tack in heel area.	X	
	Any protruding point of lasting staple, or point of tack forward of heel breast line.	1/	

Examine	Defect	Classification Major	Minor
Heel pad	Not firmly and adequately adhered to heel seat.		X
Marking	Missing, incomplete, incorrect, not applied in the specified manner, misplaced, illegible, or not specified size.		X
ANSI label	Missing, illegible, not as specified or improperly attached.		X

1/ Any protruding point of lasting staple, or point of tack found in the sample shall be cause for rejection of the lot represented. The AOL specified in 4.4.3 does not apply to tack or staple forward of heel breast line.

4.4.4 End item testing. The finished shoe shall be tested for the characteristics listed in table XII. The sample unit shall be one shoe. All test reports shall contain the individual values utilized in expressing the final result. Requirements are applicable to the sample unit. The lot shall be unacceptable if one or more sample units fail to meet any requirement specified in table XII. The sample size shall be as follows:

Lot size (pairs)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,001 and over	5

TABLE XII. End item tests

Characteristic	Requirement paragraph	Test method	Number of determinations per sample unit
Impact resistance	3.3.4.1.1	4.5.4	1
Bond strength of outsole and heel	3.7.15	4.5.5	1

* 4.4.5 Electrical testing. Every shoe shall be tested in accordance with procedure specified in 4.5.6 for conformance to the requirements specified in 3.9. Any shoe failing one or more of the electrical tests shall be rejected. The contractor shall furnish a certificate of compliance with each lot stating that every shoe has been tested and complies with the electrical requirements.

4.4.6 Packaging inspection. An examination shall be made to determine that the preservation, packing, and marking comply with the section 5 requirements. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged with the exception that it need not be closed. Examination for closure defects listed below shall be made on shipping containers fully packaged. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AOL expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and unit pack)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Any component missing, damaged, or not as specified.
Workmanship	Inadequate application of components, such as: incomplete closure of container flap, loose strapping, improper taping, or inadequate stapling. Open and non-continuous heat sealed seams and closure of polyethylene bag. Incorrectly fabricated bag. Bulged or distorted container.
Content	Number of pairs of shoes per container is more or less than required. Size indicated on shoes not the same as specified on unit pack or shipping container. 1/

1/ For this defect, one unit pack from each container in the sample shall be examined.

4.5 Methods of inspection.

4.5.1 Non-marking test. One corner of the specimen shall be drawn firmly across a sheet of white bond paper. Any resulting marking which cannot be easily removed by rubbing with the fingers shall be considered a test failure.

* 4.5.2 Cut growth after aging test. Cut growth after aging shall be determined in accordance with ASTM Method D 1052, except that the results shall be reported as percent cut growth after 25,000 cycles and a reading after each 100 percent cut growth shall not be reported. Aging shall be conducted in accordance with ASTM D 573. Specimens shall be aged 24 hours at 100° ± 1°C.

* 4.5.3 Bond strength test of outsole and midsole. The specimen shall be a combined midsole-outsole unit which has aged at least 2 days. The outsole shall be separated from the midsole for a distance of approximately 2-1/2 inches from the toe end of the specimen. A power-driven portable gage tong adhesion machine, or an approved portable testing device of equal performance shall be used. The rate of travel of the power actuated grip shall be 2 inches per minute. The machine shall be operated without any device for maintaining maximum load indication. The separated toe ends of the specimen shall be clamped in the jaws of the machine, with the jaws 2 inches apart. The specimen shall extend outward at right angles to the direction of the application of the load. The machine shall be started and the outsole and midsole shall be pulled apart for a distance of 1 inch. At that instant, the load indicated on the machine shall be read and recorded. The load shall be divided by the width of the specimen at the corresponding line of separation.

4.5.4 Impact test of steel box-toe. The impact test shall be made on a finished shoe or on the toe section of a shoe obtained by sawing through the short dimension of the shoe not less than 1/2 inch back of the box-toe. A falling weight of steel or other suitable material, weighing not less than 49-1/2 pounds nor more than 50-1/2 pounds, 1 ± 0.02 inch in diameter and with a striking face made hemispherical in shape by rounding on a 1 inch radius, equipped to drop freely in a tube or other suitable guide, shall be used. The weight shall be dropped from a distance of 18 ± 1/4 inches above the top of the box toe. As an alternate, a falling weight of 25 ± 1/4 pounds shall be dropped from a distance of 36 ± 1/2 inches. The equipment shall be assembled so that the center of the striking face of the falling weight will strike the shoe at midwidth 1/2 inch in front of the back edge of the box toe. The shoe (or toe-sections) shall be mounted on an anvil (or rigidly mounted bedplate) of steel or other suitable material weighing not less than 500 pounds, equipped with suitable guides (or clamps) to hold shoe specimen in place. In performing the test, the instantaneous clearance may be determined by placing lumps of plastic material such as wax or modeling clay inside the shoe directly below the point of impact. The plastic material should be compressed with the fingers so that it contacts both upper and lower inside surfaces of the shoe, before the test is made. When in place, the greatest horizontal dimension of the lumps should not exceed 1 inch. An approved method for impact testing is contained in American National Standard ANSI Z41.1.

portion of the sole stitching that runs

* 4.5.5 Bond strength test of outsole and heel. The specimen shall be a complete shoe which has aged at least 3 days. Starting at the back of the heel separate the heel from the outsole for a distance of 1 inch. A power driven pendulum type machine or an approved portable testing device of equal performance shall be used. The rate of travel of the power actuated grip shall be 2 inches per minute. The machine shall be operated without any device for maintaining maximum load indication. The separated ends of the heel and sole shall be clamped in the jaws of the machine. "C" clamps or other rigid clamps may be used to fasten the specimen to the jaws of the testing machine. The machine shall be started, and the outsole and heel shall be pulled apart for a distance of 1 inch. At that instant, the load indicated on the machine shall be read and recorded. The load shall be divided by the width of the specimen at the corresponding line of separation, and reported as pounds per inch of width.

4.5.6 Electrical tests.

4.5.6.1 Test equipment. The test equipment shall have a rating of not less than 1/2 KVA per shoe being simultaneously tested and the impedence value of the measuring system shall not exceed 280,000 ohms. The electrodes shall consist of a plate electrode and a suitable block shod with sponge rubber on which is placed a layer of metal foil.

4.5.6.2 Procedure. The proof test for current leakage, and breakdown voltage test shall be performed on each shoe of each lot by mounting the shoe so that the sole and heel are in contact with the plate electrode (external electrode). The internal electrode shall be made of sponge rubber cut to the shape of an insole and wrapped with metal foil and shall be placed in the shoe with the foil in contact with the insole of the shoe. A conductor from this electrode is to be brought out of the shoe so that the electrode can be connected to the test circuit. A special block (modified shoe last or similar) is then placed in the shoe on the internal electrode and a load of 75 pounds applied to the block.

* 4.5.6.3 Proof test for current leakage. The proof test voltage shall be applied starting at 0 volts and gradually and steadily raised at a rate of approximately 1,000 volts per second until 10,000 volts (rms) proof test voltage is reached (or failure occurs). The 3-minute test period shall be counted from the instant the prescribed testing voltage is reached. The proof test current shall be measured directly, by inserting a milliammeter in series with each individual shoe. The measurement shall preferrably be taken near the end of the proof voltage test period. It is customary to make high-voltage tests of this character with one end of the test circuit grounded. The milliammeter is connected in the grounded end of the test circuit with a short-circuiting device shunting the meter. This will keep the circuit closed except at the time the current measurements are being made, thus maintaining an uninterrupted ground. If the milliammeter and short circuiting device are adequately insulated, they may be used in the high voltage circuit. Leakage current exceeding 10 milliamperes shall be considered a test failure.

4.5.6.4 Breakdown voltage test. The breakdown test voltage of 14,000 volts (rms) shall be applied to the shoe by raising the test voltage from 10,000 to 14,000 volts (rms) at a rate of approximately 1,000 volts per second. The 1-minute testing period shall be counted from the instant the prescribed testing voltage is reached. Leakage current exceeding 14 milliamperes shall be considered a test failure.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each pair of mated shoes shall be inserted in a close-fitting clear polyethylene film bag having a thickness of 0.002 inch (± 20 percent tolerance). The bag shall be formed with heat sealed seams that are straight, continuous and parallel to each other and the formed edges of the bag. The final closure of the bag shall be heat sealed with the heat seal made as close as possible to the open end. The bag may be fabricated from polyethylene film tubing or sheeting. Alternatively, the final closure of the polyethylene bag may be accomplished by means of a mechanical tie (paper or plastic covered soft steel wire, aluminum band, etc.). Each bagged pair of shoes shall then be placed in a two-piece partial telescope style folding or setup paperboard box as hereinafter specified. The depth of the lid of the box shall be 1 inch.

5.1.1.1 Folding boxes. Folding boxes shall be fabricated from a kraft-lined chipoard not less than 0.032 inch thick conforming to variety 1, style III, type G, class j, subclass 2 of PPP-B-566.

5.1.1.2 Setup boxes. Setup boxes shall be fabricated from paperboard not less than 0.040 inch thick conforming to type II, variety 1, class A or D, style 4 of PPP-B-676.

* 5.1.2 Commercial. The shoes shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Twelve pairs of shoes of one size and width only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Each fiberboard container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.6. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2 Level B packing. Twelve pairs of shoes of one size and width only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636 except that the inspection shall be in accordance with 4.4.6. Toward the end of the contract or when there are less than the required amount per container of the same size, mixed sizes may be packed within the same container.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of the container specification, except that the inspection shall be in accordance with 4.4.6.

* 5.2.3 Commercial packing. Shoes, preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs and shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.3.1 Width marking. Following the nomenclature and size designation, in the same size lettering, the width shall be spelled out; i.e., "EXTRA NARROW", "NARROW", "REGULAR", "WIDE", or "EXTRA WIDE".

* 5.3.2 Labels, mixed sizes. Each shipping container packed with mixed sizes, shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches with the words "MIXED NSN's" plainly stamped or printed thereon, and under these words shall be legibly stamped or printed the correct quantity of pairs and NSN's contained therein.

6. NOTES

6.1 Intended use. The shoes are intended for use by personnel where assigned duties require work on energized circuits not exceeding 600 volts. The bottom of the shoe will afford the wearer protection against electrical hazards, however, the upper part will not, since lethal currents can readily leak through the leather. The shoes should not be worn in flooded areas where water might saturate the shoe above the welt line.

* 6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number and date of this document.
b. Sizes and widths required (see 1.2).
c. When a first article is required (see 3.2, 4.3 and 6.4).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).

6.3 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample consisting of one finished pair of shoes. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for inspection and approval of the first article.

6.5 Satra scale. The Satra scale may be obtained from the British Shoe and Allied Trade Research Association, Satra House, Kettering, Northants, England or may be obtained from Satra House's North American Agent: Bata Engineering, Batawa, Ontario, Canada.

6.6 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

6.7 Changes from previous issue. The margins of this document are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - NU
Air Force - 99

Preparing activity:

Army - GL

Project No. 8430-0325

Review activities:

Army - MD
Navy - MC
Air Force - 11, 82
DLA - CT

User activities:

Navy - OS
Air Force - 45

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE ARMY
US Army Natick Research and Development Center
Natick, Massachusetts 01760

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300
STRNC-ES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12062 WASHINGTON D. C.
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Natick Research and Development Center
ATTN: STRNC-ES
Natick, MA 01760

[redacted]

[illegible] modified polyamide hardener [illegible] conforming to the fol[illegible]

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-43897B	Shoes, Electrical Hazards Protective, Oxford, Men's

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 APR 83